ENGR-UH 2113 *Muhammad Ibrahim*

Introduction to Manufacturing *NetID: mi2171*

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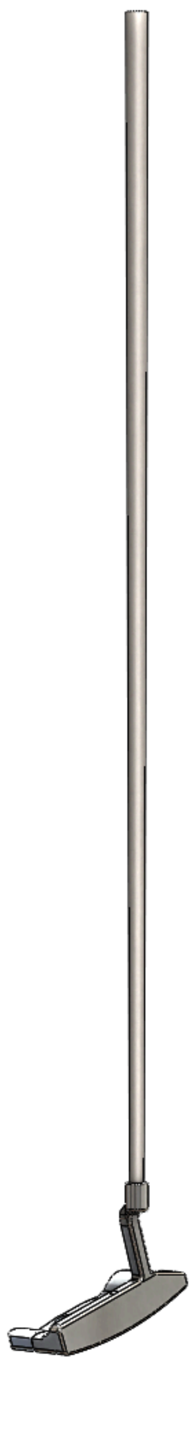
Designing a Modular, Homologated Putter



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Iterative Design

Both prior and post midterm submission

From the very beginning, the putter's development underwent multiple refinements to address both functional and aesthetic considerations. In the first iteration, I aimed for a compact 11 cm face length to enhance control, pairing it with a slim plumber neck that featured decorative grooves for weight reduction and visual appeal. This initial model also included a circular toe weight, screwed into the putter head to shift the center of mass outward. While increasing the mass moment of inertia and delaying face rotation, this approach proved problematic for reliably squaring the face at impact. Recognizing these challenges, I moved on to a second iteration that incorporated a 1.1 scale-up of the entire putter head, a more robust and flush plumber neck, and a shift from inlays to magnets for weight attachment. By placing the center of gravity behind the shaft and providing modularly attached weights, the second design allowed for a more balanced stroke and easier customization. Ultimately, the final iteration refined the putter further by adding a grip to improve accessibility, placing magnets directly in the putter head to better secure the weights, and enlarging the lettering on the sole for easier cleaning and maintenance. Each step of this iterative journey served to identify shortcomings, test potential solutions, and implement tangible improvements.

For the central weight assembly, we started with 629.78 grams, then added 2.36 grams for the magnets (bringing it to 632.14 grams), and finally added 70 grams for the grip, which gives a total mass of 702.14 grams. For the toe weight assembly, we started with 623.27 grams, then added 2.36 grams for the magnets (resulting in 625.63 grams), and then added 70 grams for the grip, for a total of 695.63 grams.

Figure 1: Design 1

Figure 2: Design 2 (midterm submission)

Figure 3: Design 2 (midterm submission)

Figure 4: Design 3

Figure 5: Design 3 with alignment weights

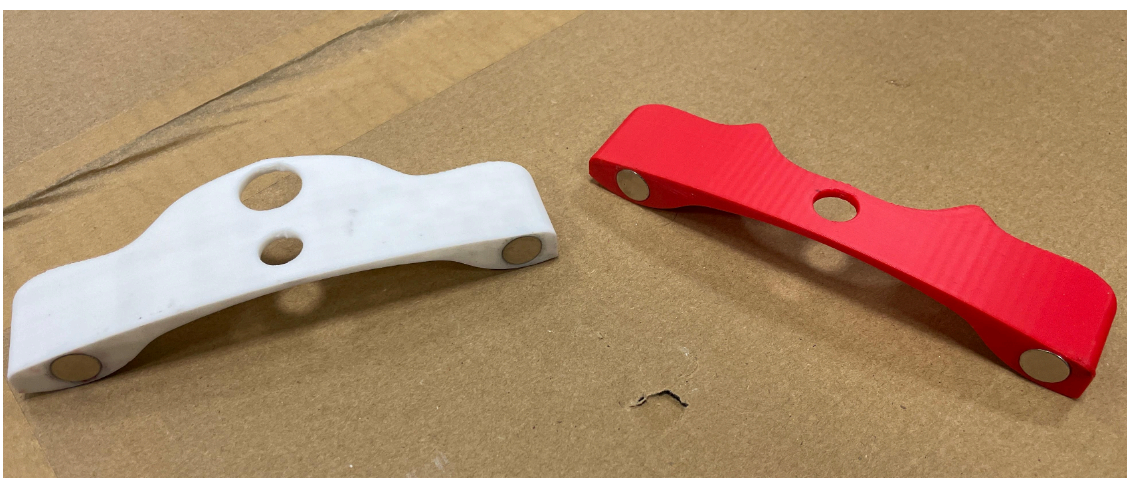

Figure 6: Weights with magnets

Design Improvements

Post Midterm Submission

A key improvement evident in the progression from the first to the second iteration was the decision to enlarge the putter head slightly, which improved both stability and feel by moving the center of gravity to a more advantageous position behind the shaft. Replacing the single screw-in toe weight with a pair of magnetically attached weights—one central and one in the toe—allowed for a versatile weighting system that better accommodated different putting strokes. This modular approach not only offered ease of adjustment but also helped avoid undue stress concentrations in the head. Furthermore, making the plumber neck thicker and more seamlessly integrated with the putter head reduced vibration and added stability during the stroke. For the final iteration, I introduced a grip on the shaft, recognizing the importance of comfort and accessibility for all golfers, especially those who experience joint discomfort. Alongside this, magnets were added to both the putter head and the weights to ensure that the weights remained firmly aligned without the need for potentially fragile alignment grooves. This cleaner, more durable solution provided long-term reliability while reinforcing the putter's customization potential. Lastly, enlarging the inscribed lettering on the sole offered a practical benefit for easier maintenance, underscoring the attention to detail in every aspect of the design.

Figure 7: Large Diameter Golf Grip [1]

Challenges

Both prior and post midterm submission

Despite these advancements, I encountered a variety of challenges that required careful consideration and creative solutions. Initially, sourcing titanium for the weights proved difficult, leading me to use aluminum as a replacement. While aluminum provided an acceptable substitute, it introduced its own considerations for balancing and durability. Additionally, alignment grooves were inscribed instead of fully milled to avoid overly thin sections that risked fractures and premature wear. The plumber neck had to be smoothened to eliminate high stress points and reduce the likelihood of failures over time. In parallel, the reliance on magnets during the second iteration addressed issues with weight shifting, but I found that magnetizing only the weights caused minor movement during the stroke, prompting me to install magnets in the putter head as well.

Another significant hurdle was the actual prototyping process itself. With limited manufacturing machinery, insufficient time, and restricted access to raw materials, I resorted to 3D printing the putter head, weights, and shaft. Although 3D printing was a practical solution under these constraints, it highlighted the need for more precise tolerances and robust materials for a final prototype. The prototype also included a 3D-printed grip, and the original 3D-printed shaft proved too flexible due to the inherent nature of layered printing. To address this, I used a tougher filament and increased the infill density, which reduced shaft flexibility. However, there would be no way to match the sturdiness of a steel shaft. Each of these obstacles ultimately guided the putter's evolution, influencing design changes and reinforcing the importance of thorough testing and adaptation at every stage.

Final Costing

While the solidworks costing algorithm did not produce an increase in costs, there have been some reasonable estimates made for the putter head changes.

In a lot size of 50 units, the ***original base costs*** per unit were: putter head at \$220.10, shaft (Apollo 36" Steel) at \$19.44, 20 magnets at \$1.578, toe weight at \$36.64, and central weight at \$32.74, yielding a subtotal of \$310.498. After a 1% increase to the putter head (\$222.30) and switching to 22 magnets (\$1.736), the new base subtotal becomes \$312.856, and adding the 20% assembly fee (\$62.57) brings it to \$375.43. Including the grip, which costs 189 AED (approximately \$51.50), the ***final per-unit cost*** now totals \$426.93—an increase of \$54.33 over the previous \$372.60.

For single-piece production (lot size of 1), the ***original costs*** were: putter head (\$1,410.55), toe weight (\$455.65), central weight (\$437.33), and shaft (\$344.22) for a subtotal of \$2,647.75, plus \$2,000 in mold costs for a total of \$4,647.75. A 20% assembly fee (\$929.55) raises this to \$5,577.30, a 12% new customer fee (\$669.28) brings it to \$6,246.58, and a 10% miscellaneous cost (\$624.66) results in the original total of \$6,871.24. With the 1% putter head increase (\$1,424.66), two additional magnets (22 total, adding about \$1.74), and the \$51.50 grip, the revised part costs become \$2,715.10, plus the same \$2,000 for molds, yielding \$4,715.10. A 20% assembly fee (\$943.02) then pushes it to \$5,658.12, followed by a 12% new customer fee (\$678.97) for \$6,337.09, and finally a 10% miscellaneous fee (\$633.71) for a ***final single unit cost*** grand total of approximately \$6,970.80—\$99.56 more than the original \$6,871.24.

Production Type	Initial Cost (USD)	Latest Cost (USD)	Difference (USD)
Single Production (Lot = 1)	6,871.24	6,970.80	99.56
50-Lot (Per Unit)	372.60	426.93	54.33

*Table 1: Comparison of **Initial** and **Final** Costs for Single Production and 50-Lot (Per-Unit) Costs*

Item	Quantity	Unit Cost (USD)	Subtotal (USD)
Putter Head (1% increase)	1	1,424.66	1,424.66
Toe Weight	1	455.65	455.65
Central Weight	1	437.33	437.33
Shaft (if manufactured)	1	344.22	344.22
Magnets (22 total)	22	0.0791	1.74
Grip (189 AED ≈ $51.50)	1	51.50	51.50
		Subtotal (Parts)	2,715.10
		Mold Costs (2 × $1,000 each)	2,000.00
		Subtotal (Parts + Molds)	4,715.10
		20% Assembly Fee	943.02
		Subtotal	5,658.12
		12% New Customer Fee	678.97
		Subtotal	6,337.09
		10% Miscellaneous Cost	633.71
		Grand Total	6,970.80

*Table 2: Bill of Materials for Single-Piece Production (Lot Size = 1) (**Final Cost**)*

Item	Quantity	Unit Cost (USD)	Subtotal (USD)
Putter Head (1% increase)	1	222.30	222.30
Shaft (Apollo 36" Steel)	1	19.44	19.44
Magnets (22 total)	22	0.0789	1.736
Toe Weight	1	36.64	36.64
Central Weight	1	32.74	32.74
		Subtotal (Parts)	312.856
		20% Assembly Fee	62.57
		Subtotal	375.43
		Grip (189 AED ≈ $51.50)	51.50
		Grand Total (Per Unit)	426.93
		Grand Total (50 Units)	21,346.50

*Table 3: Bill of Materials for One Unit in a 50-Unit Lot (**Final Costs**)*

Final Prototype

Figure 8: Complete Prototype

Figure 9: Complete Prototype

Figure 10: Complete Prototype

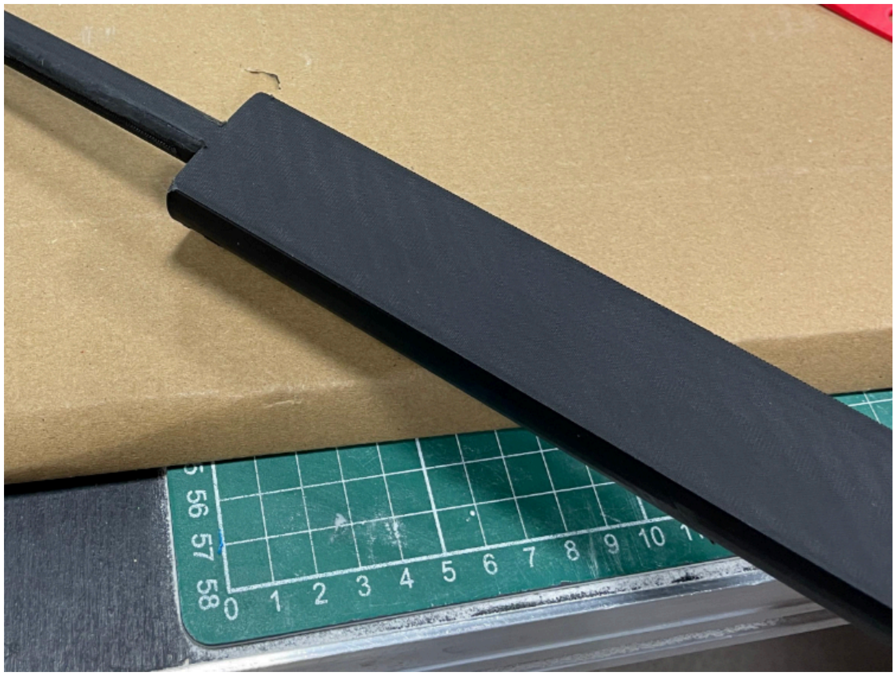

Figure 11: Ergonomic Grip

References

[1] EGolfMegastore. (n.d.). *Superstroke Legacy 2.0 Counter Core Putter Grip (Red)*.